Lecture 8

Factoring State Machines

Factoring State Machines

- Break complex FSMs into smaller interacting FSMs
- Example: Modify traffic light controller to have Parade Mode.
  - Two more inputs: *P*, *R*
  - When ***P* = 1**, enter Parade Mode & Bravado Blvd light stays green
  - When ***R* = 1**, leave Parade Mode

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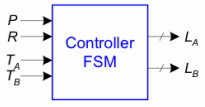
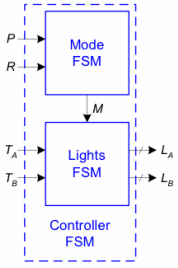
Parade FSM

Unfactored FSM



Factored FSM



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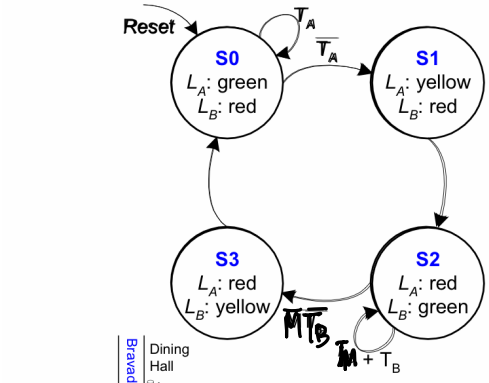
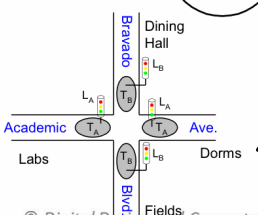
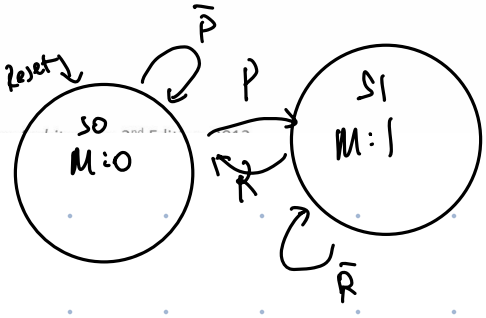
Factored FSM



Lights FSM





Mode FSM.

Review: FSM Design Procedure

1. Identify **inputs** and **outputs**
2. Sketch **state transition diagram**
3. Write **state transition table**
4. Select **state(input, output) encodings**
5. For Moore machine:
   1. Rewrite state transition table with state **encodings**
   2. Write **output table**
6. For a Mealy machine:
   1. Rewrite combined state transition and output table with state **encodings**
7. Write **Boolean equations** for next state and output logic
8. Sketch the circuit **schematic**

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A Weird Weather Light System

I have a weird weather light system. The system monitors weathers of four cities, New Haven, New York, Boston, and Springfield. There are two lights, a green light and a blue light. The green light will be on if one day ago three or four of the cities were sunny, otherwise, off. The blue light will be on only if two days ago there was only one city was sunny.

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(1) Input : # sunny city (0, 1, 2, 3, 4)

Clock, reset.

3, 4

1

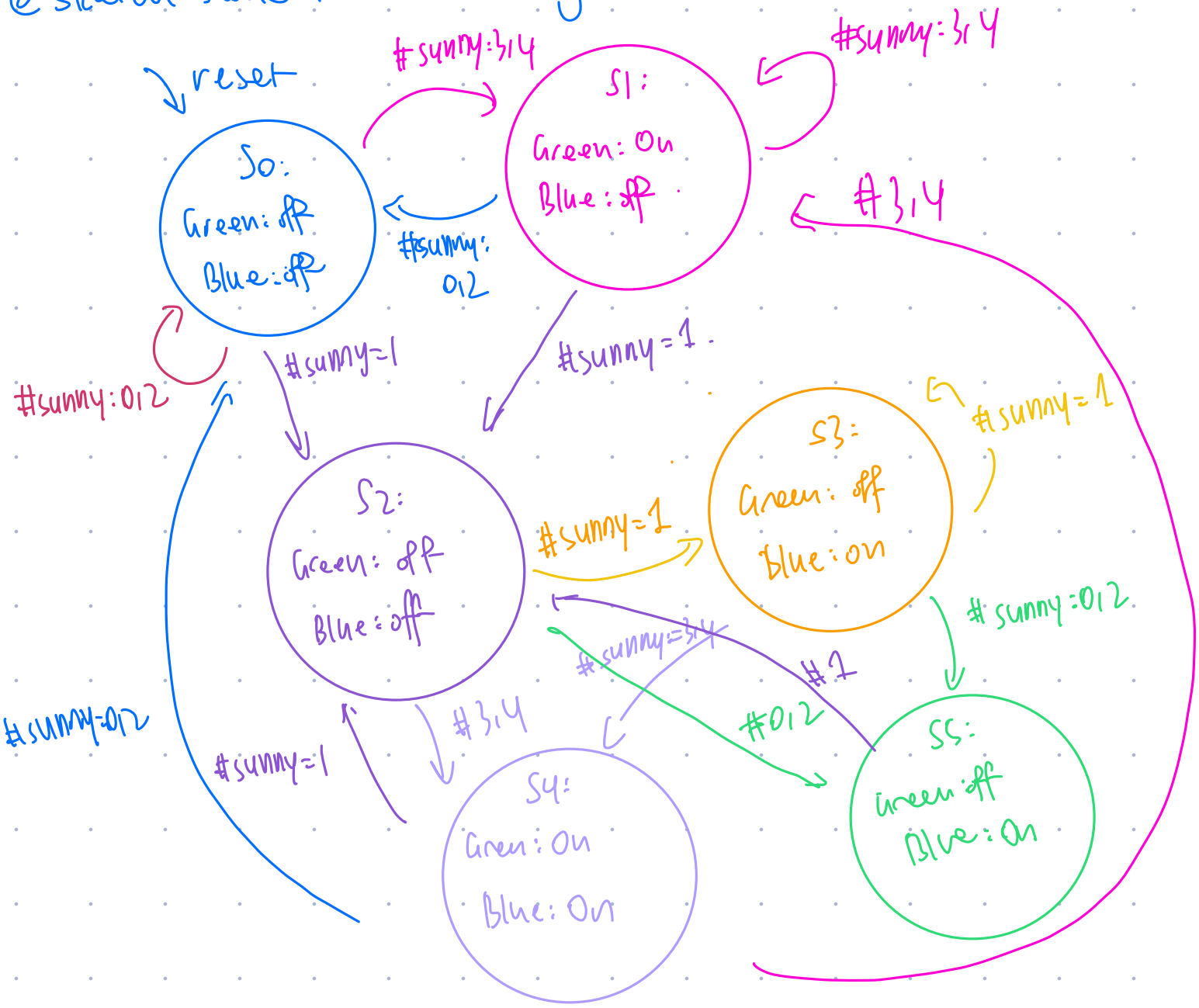
0, 2

can be grouped into 3 categories.

Output: Green : On, off

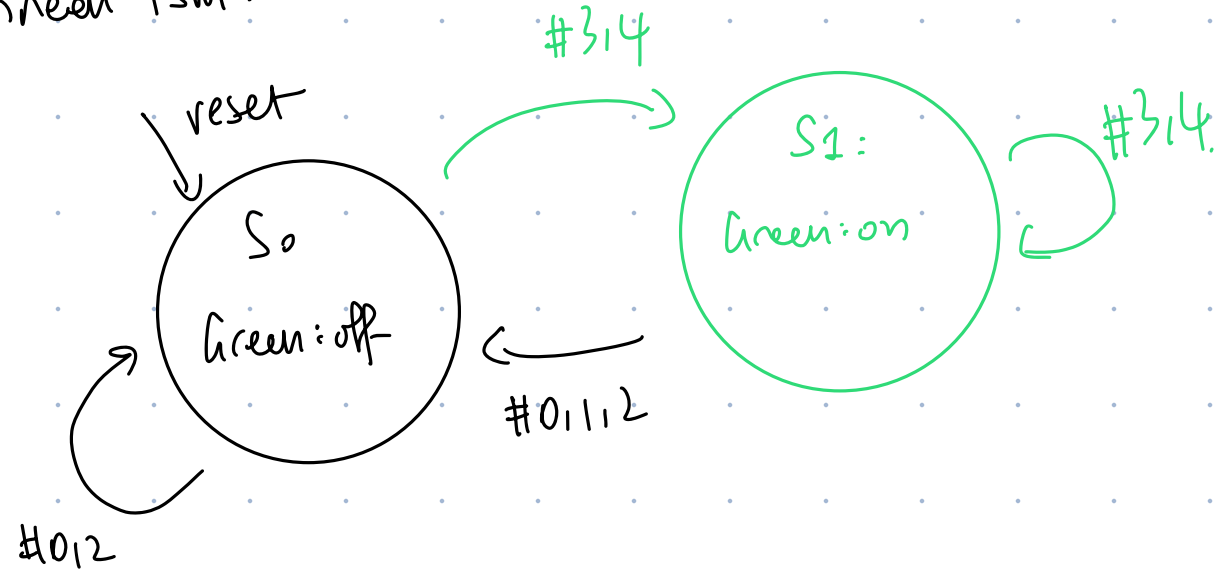
Blue : On, off

② sketch state transition diagram:

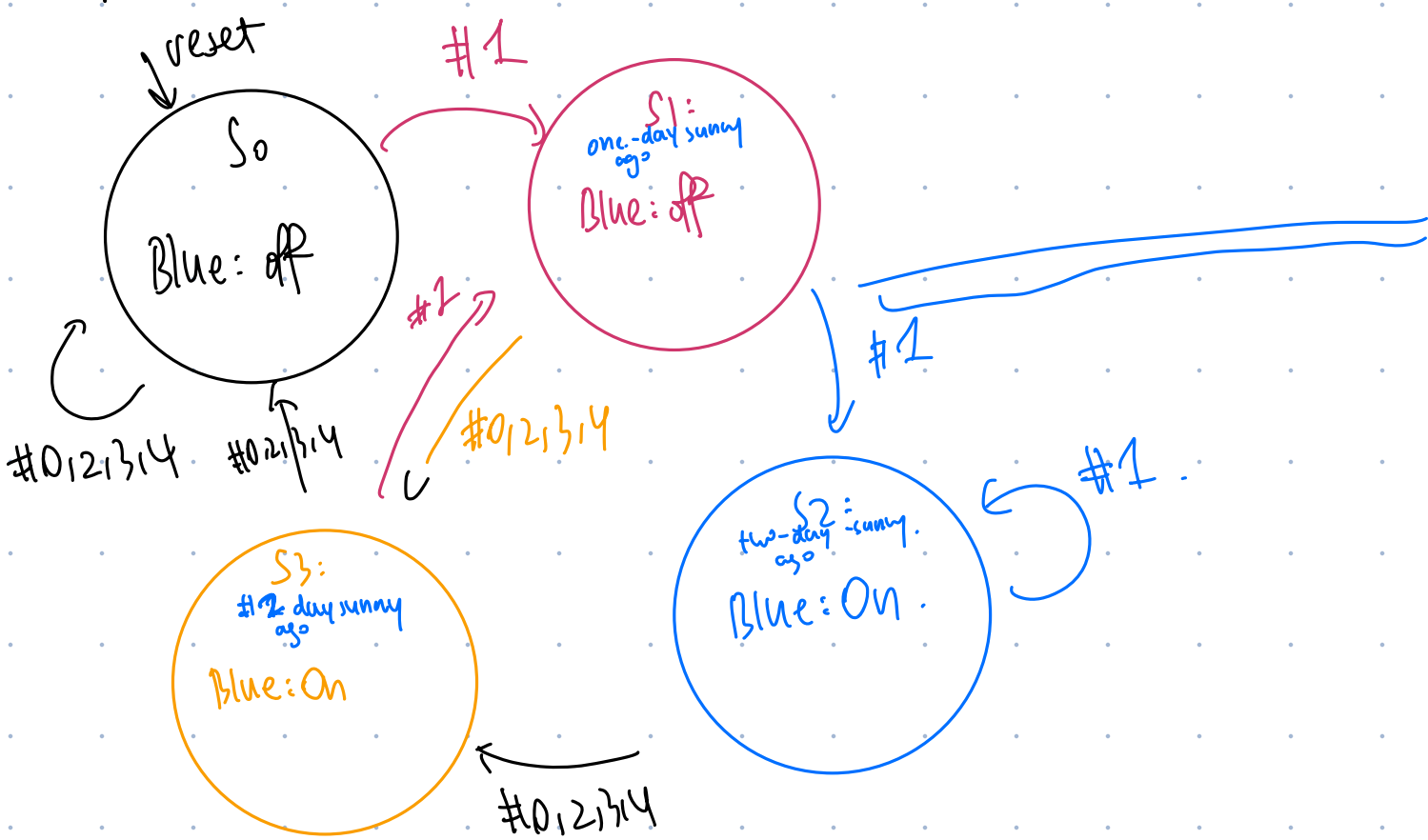


this is an unfactored FSM.

Green FSM.



Blue light.



Green light

anoding

Current State	input ①	next state
S0 / 0	0, 1, 2 / $\bar{W}$	S0 / 0
S0 / 0	3, 4 / W	S1 / 1
S1 / 1	0, 1, 2 / $\bar{W}$	S0 / 0
S1 / 1	3, 4 / W	S1 / 1

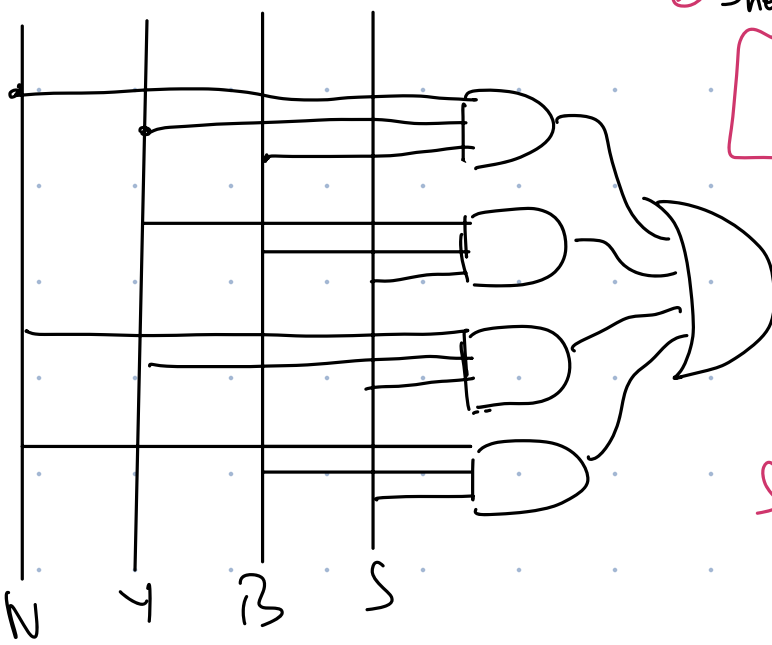
?

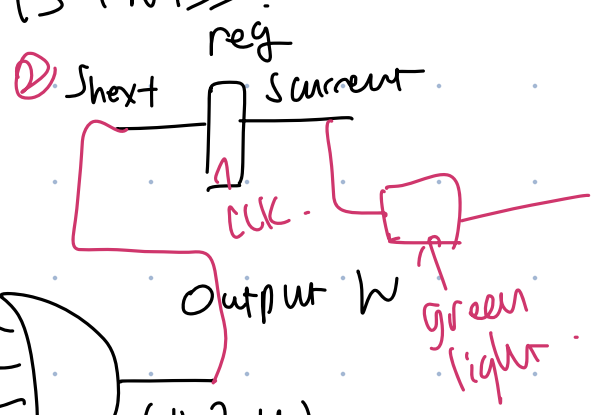
NY \ BS	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	1	1	1
10	0	0	1	0

#3,4

0 ⇒ 1

$$Y = NYB + YBS + NYS + NBS$$





③

$S_{next} = \overline{S_{current}}\, W + S_{current}\, W = W$